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Simultaneous Combustion Instability and Emissions Control Using Air and Fuel Modulation

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Abstract

We study experimentally the impact of static and modulated transverse jet injected near the flame anchoring zone on the stability and emissions in a premixed backward facing step combustor. The combustor shows strong pressure oscillations at 38 *Hz.* Parametric studies using multiple sensors indicate that flame-vortex interactions are responsible for the heat release dynamics under unstable conditions. We demonstrate that with sufficient momentum, a static air jet injected perpendicular to the incoming flow near the step favorably modifies the flame-vortex interactions and reduces the pressure oscillations by up to 14 *dB* (OASPL), and the NOx emission by an order of magnitude. Adding small amount of hydrogen to the primary fuel further reduces the pressure oscillations. We also investigate the impact of the transverse jet modulation. Result shows that one can reduce the amount of air necessary for stabilization by 50% from the static case by using closed-loop control and jet modulation.

1. Introduction

Continuous combustion systems common in propulsion and power generation applications are susceptible to thermoacoustic instability, which occurs under lean burn conditions close to the flammability where most emissions and efficiency benefits are achieved, and near stoichiometry where high power density is the objective. Resonance between the acoustic field of the combustion chamber and the heat-release rate has been shown under these conditions. Pressure oscillations inside the combustor cause fluctuations in the heat-release rate through several mechanisms, including equivalence ratio fluctuations [1,2], flame surface oscillation [3], flame-vortex interactions [4-8], either individually or simultaneously [9]. Under certain conditions the pressure and heat-release fluctuations are in phase, causing an unstable feedback between the hear release and the acoustics [10].

Active control of combustion dynamics is an attractive approach for achieving desired performance goals, such as reduced overall sound pressure level, reduced emissions, and increased efficiency. Typically, active combustion control has been implemented using pressure or heat release rate sensors, and an actuator that modulates the fuel [11-14] or air streams [15]. Fuel modulation suffers from a possible emission's penalty associated with fuel injection into the combustion zone, where maximum authority is expected. Moreover, modulating the primary fuel in the upstream may be ineffective if the heat-release dynamics are governed by flame-vortex interactions rather than equivalence ratio oscillations.

In this paper, we first investigate the thermoacoustic instability mechanism in an 80 *kW* premixed backward-facing step combustor in order to determine the appropriate actuation technique. Parametric studies and flame images from various sensors confirm that the dominant mechanism is flame-vortex interactions throughout the entire range of operating conditions.

Secondly, we examine the use of steady and modulated transverse jets, injected near the combustion zone, as means of simultaneously suppressing the pressure oscillations and reducing NOx. We show that the transverse jet, under certain conditions, favorably modifies flame-vortex interactions, suppressing pressure fluctuations by up to 14 *dB* with an order of magnitude reduction in NO emissions. Under different conditions, using a transverse air jet near the lean flammability limit makes the flame more susceptible to blow out. Next, we experiment with the addition of small amounts of hydrogen in the primary fuel and in the air jet. It has been shown that adding a small amount of hydrogen to other hydrocarbons can extend

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their flammability limit and increase their resistance to flame extinction [16-21].

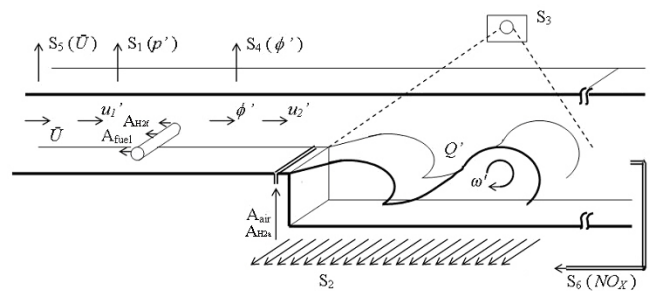
In Section 2, the experimental apparatus, the sensors and actuators are described. In Section 3, we discuss an overall combustion dynamics model that describes the interactions between the heat-release dynamics, the acoustics, and the hydrodynamics. A preliminary parametric study that sheds light on the flame-vortex interactions and their impact on instability is presented in this section. We conclude with open-loop and closed-loop control results using air jet injection and H2 addition in Section 4.

2. Combustor Description

The experimental tests are conducted in a backward-facing step combustor, in which a sudden expansion is used to anchor the flame, as shown in Fig. 1. The combustor consists of a stainless steel duct with a rectangular cross section. The duct is 160 *mm* wide. At the test section, its height expands suddenly from 20 *mm* to 40 *mm* forming a backward-facing step. The overall length of the combustor is 1.9 *m*, with the step located at the middle. The test section is equipped with quartz viewing windows for optical access. The air inlet is choked. The exhaust goes through a sudden expansion at the end of the tunnel with an expansion ratio of 1 to 5. For more detail, we refer the reader to Ref. [22]. The sensors and actuators are positioned as indicated in Fig. 1, where p' is pressure fluctuations measured at 74 *cm* upstream of the step, $\overline{U}$ is the mean air velocity in the upstream section, u_1' and u_2' are the velocity fluctuations at the point of fuel injection and the step, respectively, Q' is the fluctuation in the heat release rate, ω' is vorticity fluctuations downstream the step, ϕ' is equivalence ratio fluctuations measured at 12 *cm* upstream of the step, and NOx is measured 62 *cm* downstream the step. A_{fuel} and A_{air} are the locations for fuel and air forcing used. A_{H2f} and A_{H2a} are the locations for H2 forcing in the upstream and at the air injection slot, all used for control. Figure 2 is a photograph of the test section of the combustor with key sensors and actuators identified by the symbols defined above.

2.1 Sensors

Here, we describe the sensors used to identify the state. Pressure fluctuation is measured using a Kistler pressure sensor (S_1). CH* chemiluminescence, which can be related to heat-release fluctuation [23,24], is measured using a linear photodiode array (S_2). The 2D flame structure is visualized using a high speed camera (S_3). Equivalence ratio fluctuations are characterized using a He-Ne laser and an In-As detector arrangement (S_4) [25]. The air mass flow is measured using a Sierra Instruments 780S-Flat-Trak (S_5). NOx is measured using the Thermo Environmental Instruments Model 42C High Level NO-NO_2-NO_X Analyzer (S_6). The pressure, equivalence ratio, air mass flow and NOx measurements are logged at 500 Hz into a Pentium III computer equipped with a dSPACE data acquisition board. The linear photodiode array voltages are captured by a Pentium IV computer equipped with National Instruments data acquisition hardware. The overall sampling rate from the photodiodes is 64 kHz, and each snapshot is captured at 500 Hz.



Sensors		Variables	
S_1	Pressure Sensor (p')	p'	Pressure Fluctuations
S_2	Linear Photodiode 128-array (Q')	Q'	Heat Release Fluctuations
S_3	High Speed Camera	ω'	Vorticity
S_4	Equivalence Ratio Sensor (ϕ')	ϕ'	Equivalence Ratio Fluctuations
S_5	Air Flow Meter ($\bar{U}$)	$\bar{U}$	Mean Air Flow Rate
S_6	Emissions (NO_X)	u_1'	Velocity Perturbations at the step
Actuators			
A_{air}	Air Forcing	u_2'	Velocity Perturbations at the fuel bar
A_{fuel}	Fuel Forcing		
A_{H2f}	H2 Forcing with the Fuel		
A_{H2a}	H2 Forcing with Air		

Fig. 1 Schematic diagram of the backward-facing step combustor with various sensors and actuators

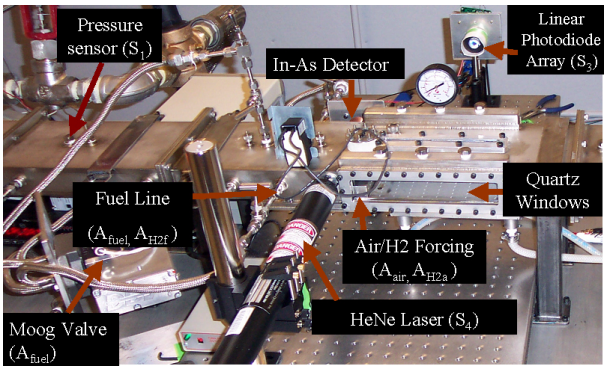


Fig. 2 Photograph of the backward-facing step combustor and some of the sensors used.

2.2 Actuators

As indicated in Fig. 1, the primary fuel injection occurs several step heights upstream of the step through a bar aligned in the spanwise direction, with multiple fuel jets pointing upstream to improve the streamwise mixing. While the primary fuel can be modulated at an order of magnitude higher than the fundamental unstable frequency, downstream mixing between the injection location and the flame anchoring position reduces the fuel modulation authority. Moreover, actuation far upstream introduces a time delay between fuel modulation and the combustion event. Alternatively, air forcing through a spanwise slot located 12 *mm* upstream of the step occurs much closer to the dump plane, and can deliver higher authority. Furthermore, adding additional air through the slot near the step, besides its significant fluid dynamic effect, tends to decrease the local temperature, which is favorable for NO_x reduction but may endanger flame stability. To overcome this limitation, a small amount of hydrogen maybe introduced either with the primary fuel or at the air forcing location. The following provides detailed description of the air and fuel actuation. Control results with air actuation will be shown in Section 4.

Air Forcing(A_{air}) The air jet, introduced through a spanwise 2 *mm* slot located 12 *mm* upstream the step, is actuated using four Dynamco D1B2204 Dash 1 direct solenoid poppet air valves. Each valve can supply 1.8 *g/s* of air when driven by 689 kPa. The valves are connected to a plenum beneath the slot.

Primary Fuel Forcing (A_{fuel}) Propane modulation is accomplished using a Moog D633-7315 AIC Direct Drive Valve (DDV), which has its own built-in feedback loop to ensure the spool position using an LVDT with a bandwidth of 450Hz.

H2 (A_{H2f} & A_{H2a}) To extend the flammability limit and increase the resistance of the flame to extinction, hydrogen is delivered at 0-9 *mg/s* through a Sierra Instrument mass flow controller. Hydrogen is either mixed with propane and introduced through the fuel bar, A_{H2f}, or mixed with the air jet and introduced through the slot, A_{H2a}.

3. Modeling of the Combustion Dynamics

Figure 3 shows schematically how various flow, combustion and acoustics mechanisms interact, and their contributions to the combustion dynamics and emissions. This model features the following:

1. The feedback mechanism that contributes to combustion dynamics, which couples heat-release dynamics and acoustics.
2. The heat-release perturbations that arise due to fluctuations in the equivalence ratio, ϕ', and hydrodynamic perturbations, ω'.

Optimal actuation for control depends on the mechanism leading to combustion instability. For example, if equivalence ratio fluctuations play a significant role, upstream fuel modulation may work best. On the other hand, if flame-vortex interaction is the governing mechanism, hydrodynamic-based actuation near the step, such as spanwise air injection, would be more effective. A parametric study of the impact of the inlet conditions and the hydrodynamics on the combustion instability is shown in the following sections.

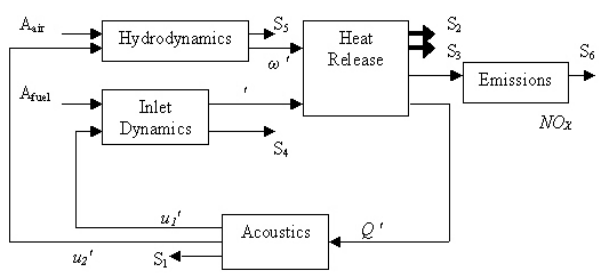


Fig. 3 Schematic diagram of the acoustics, hydrodynamics and heat release dynamics and their interactions.

3.1 The Role of the Inlet Dynamics (Mixture Inhomogeneity)

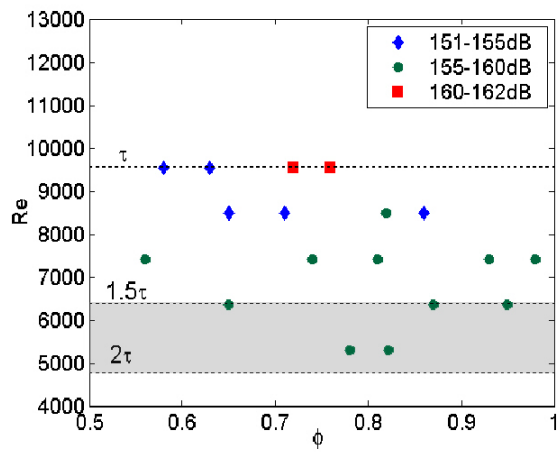
We first examine the role of fuel/air mixture inhomogeneity, ϕ', in the combustion dynamics. Equivalence ratio fluctuation occurs due to velocity perturbations, u_1', at the point of fuel injection

according to $\phi' = \bar{\phi}\dfrac{1}{1+\dfrac{u_1'}{\bar{U}}}$ (if $u_1'/\bar{U}$ <1). ϕ' appears at the burning zone following a convective transport time delay, τ_{tot}, determined by the mean mixture velocity, $\bar{U}$, and the distance from the point of fuel injection to the burning zone, L, $\tau_{tot} = \dfrac{L}{\bar{U}}$ [1,2]. Fluctuations in fuel/air mixture produce heat release fluctuations, Q', proportional to ϕ'.

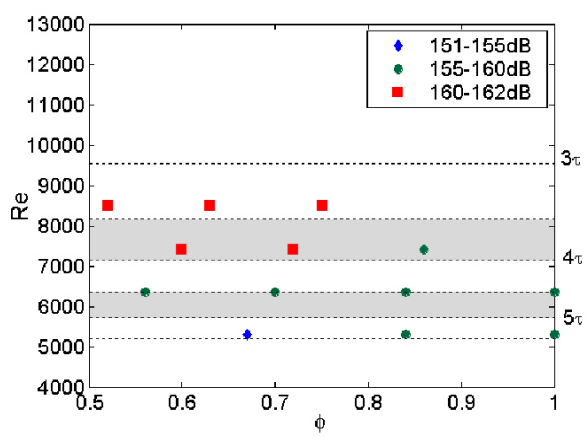
The convective delay from the point of fuel injection to the burning zone determines the phase between ϕ' and u_1' (and hence p'). The time delay can be modified by changing the fuel injection location. To determine the impact of the mixture inhomogeneity, or ϕ' on stability, the location of the fuel bar was varied and the pressure was measured at various air and fuel flow rates. Moving the fuel injection further upstream has the following effects: 1) the convective time delay is increased, and 2) the amplitude of mixture inhomogeneity is decreased due to diffusion and reduction in u_1' (the fundamental acoustic mode in the combustor is a quarter-wave mode with a closed end upstream and an open end downstream). Note that the hydrodynamic characteristics do not change by changing the fuel injection location.

We now consider two cases: I and II, where the fuel bar was located 17.5 *cm* and 35 *cm* upstream from the step, respectively. In both cases, the uncontrolled combustor was operated at different air-flow rates. The resulting stability maps are shown in Fig. 4, which depicts the effect of varying the equivalence ratio and Reynolds number on the overall sound pressure level (OASPL). In all these experiments, the excited acoustic mode frequency, f_{ac}, was 38 *Hz*. The convective time delay from the fuel injection point to the step, τ_{tot}, is shown in the figure as multiples of τ, where τ is the acoustic time scale, $1/f_{ac}$ = 26.3 *ms*. It is interesting to note that strong instability (over 160dB) occurs at or near $n\tau$ where *n* is an integer. If the instability was controlled by the mixture inhomogeneity only, a stability band should appear in these maps as the time delay is varied as follows: stable zones when the transport delay is between nτ and (n+0.5)τ and unstable zones when the delay is between (n+0.5)τ and (n+1)τ [1,2]. Changes in the pressure fluctuations were relatively small during these transitions, and data scatter was almost uniform in both regimes, indicating that the equivalence ratio oscillation is not the primary mechanism of heat release oscillations, although it may have a secondary role, as seen next.

We further investigated the extent to which the mixture inhomogeneity contributes to combustion instability. Equivalence ratio oscillations (S_4) and the corresponding pressure signal (S_1) are compared in Cases I and II at two different operating conditions. First, at Re = 5300 and ϕ = 0.85, as shown in Fig. 5a, negligible equivalence ratio fluctuations are observed in Case II due to streamwise diffusion, while significant equivalence ratio fluctuations are observed in Case I. The pressure amplitude is higher in Case I indicating that the equivalence ratio fluctuation increases the pressure oscillations at this operating condition. Note that Case I is in the band between (n+0.5)τ and (n+1)τ. On the other hand, even without equivalence ratio fluctuations, pressure oscillations are still significant in Case II.



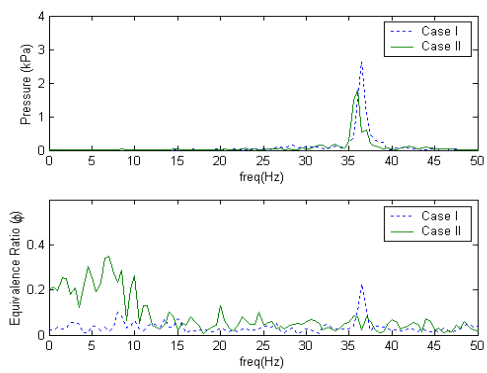
(a) Case I (fuel bar at 17.5 *cm*)



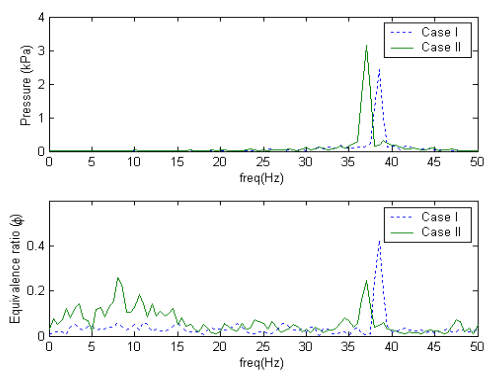
(b) Case II (fuel bar at 35 *cm*)

Fig. 4 Stability maps in Cases I and II. Values indicate the overall sound pressure level (OASPL)

The power spectra obtained at a second operating condition, at Re = 6300 and ϕ = 0.85, are shown in Fig. 5b. This condition corresponds to a band between nτ and (n+0.5)τ in both cases, where equivalence ratio oscillations may damp out pressure oscillations. While significant equivalence ratio fluctuations are present, Case II has lower values since the fuel bar is positioned further upstream, similar to the Re=5300 experiment. Since the equivalence ratio fluctuations, under these conditions, provide negative feedback, Case I exhibits a lower pressure amplitude. However, both cases show significant pressure oscillations, indicating that the essential mechanism here is not mixture inhomogeneity.



(a) Re=5300, ϕ=0.85



(b) Re=6300, ϕ=0.85

Fig. 5 Pressure (S_1) and equivalence ratio (S_4) frequency spectra at (a) Re=5300 and ϕ=0.85 and (b) Re=6300, ϕ=0.85 in Cases I and II.

3.2 The Role of Hydrodynamics and Flame-Vortex Interactions.

Velocity perturbations can cause repeated shedding of vortical structures at the step. This, in turn, causes the flame shape to change, thereby generating large amplitude of heat release fluctuation. This vortex shedding has a preferred frequency quantified by a Strouhal number $St = f_H H / \overline{U} = O(0.1)$ where H is the step height and f_H is the frequency in *Hz* [5]. If the preferred vortex shedding frequency is near the acoustic mode frequency, f_{ac}, the system is more prone to combustion instability.

The strong dependency of combustion instability on the Reynolds number or the mean velocity described above suggests that the primary mechanism causing combustion instability is the hydrodynamics and its interaction with flame. As mentioned before, hydrodynamics has a preferred frequency around $f_H = O(0.1\overline{U} / H)$. Interestingly, strong instability occurs above Re=9500 in Case I, and Re=7400 in Case II, as shown in Fig. 4, in which the preferred frequency is f_H =33Hz and 25Hz, respectively, both of which are close to the acoustic frequency, f_{ac} =38Hz. High speed CCD images in Fig. 6, captured at 500 *frames/s*, show strong flame–vortex interactions with no noticeable difference in the flame structure between the two cases. These images confirm the role of flame-vortex interactions in the combustion instability mechanism.

4. Control of Combustion Instability and Emissions

In this section, we show open and closed-loop control results using the air jet and H2 addition, in order to simultaneously reduce pressure oscillations and NOx emissions. The fuel bar is located 35 *cm* upstream of the step. In Section 4.1, we consider a static air jet (A_{air}) near the step and H2 addition to the primary fuel (A_{H2f}). In Section 4.2, we show the impact of modulating the air jet (A_{air}) when it is mixed with hydrogen (A_{H2a}). Primary fuel modulations (A_{fuel}) at the fuel bar were also attempted, but, as expected from the results of Section 3, its impact on combustion instability was negligible.

4.1 Static Air (A_{air}) and H2 injection (A_{H2f})

The effect of a steady transverse air jet (A_{air}), introduced a short distance upstream of the step, on the pressure fluctuations, and the impact of hydrogen addition to the primary fuel (A_{H2f}) in the presence of the transverse air jet as well are now examined. In these tests, the Reynolds number was 6300, with a corresponding average velocity, $\overline{U}$ = 4.4 *m/s*. The primary fuel supply was fixed. The transverse jet air

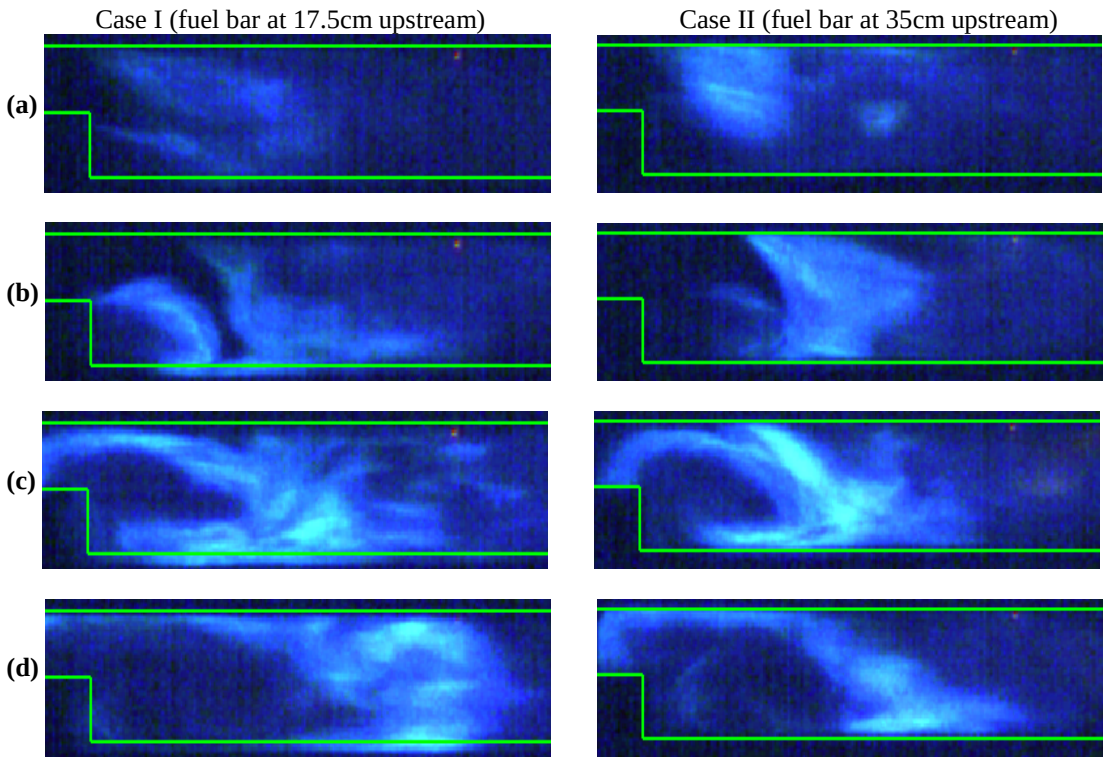


Fig. 6 High speed CCD images of the flame (Blue) in the backward-facing step combustor (green) at Re=6300 and ϕ=0.85, in Cases I and II at different moments in the cycle. (a) p'=0 and dropping, (b) minimum p', (c) p'=0 and rising, and (d) maximum p'.

velocity, u_j, was varied from 0-8.4 *m/s*. The hydrogen addition was at 1.6 and 3.2 % of the primary fuel, based on the lower heating value. The baseline equivalence ratio, ϕ=0.57. The overall equivalence ratio ranged from 0.58 to 0.47, depending on the amount of air jet and H2.

Figure 7 shows OASPL as a function of $u_j/\overline{U}$, at different H2 flow rates. For all cases, the pressure oscillations first increased for $u_j/\overline{U}$ < 1, where they reached a maximum. Above this value, the OASPL dropped rapidly. Without hydrogen, the flame blows out for $u_j/\overline{U}$ <1.2, however, with $u_j/\overline{U}$ >1.3 the flame became sustainable again, with pressure reduction of up to 14 *dB* from the baseline case. With air addition, the overall equivalence ratio was mostly below the nominal flammability limit of propane/air mixture. Moreover, a relatively large air flow rate in the transverse jet was necessary to reduce the pressure fluctuations and stabilize a flame near the step at yet lower equivalence ratio. Thus, it appears that the impact of the transverse jet is aerodynamic. Studies of simple jet in cross flow show that at small velocity ratios, the jet hardly penetrates into the stream, while as the jet velocity increases, the jet penetration is pronounced and its impact is sustained several jet widths downstream [26]. At high velocity ratios, the jet deflects the primary stream upwards, keeping it away from the lower walls. We speculate that a similar mechanism is in action here. By deflecting the stream away from the lower wall, less heat is lost to the lower wall and combustion is more adiabatic. Since the transverse jet reduces heat loss, and lowering the heat loss promotes burning of lean to very lean mixtures [27], the air jet enables burning at a lower overall equivalence ratio.

Figure 7 shows that adding H2 to the primary air stream, in the presence of the transverse jet, has a weak impact on the pressure level before the point of maximum pressure oscillation is reached. Beyond this point, the effect of H2 addition is to support a more stable flame, lowering pressure oscillation and preventing blowout. Moreover, adding hydrogen enables burning over the entire range of the jet mass flow rate (we no longer see blowout around

$u_j / \overline{U}$ =1). This is consistent with previous observations that doping hydrocarbons with small amounts of H2 extends their flammability limits and reduces their tendencies to extinguish under the influence of strong strains [16-21]. It is interesting to observe that while 1.6 % H2, measured as a percentage of the primary fuel based on the lower heating value, improves flame stability, doubling that amount did not prove to be more effective.

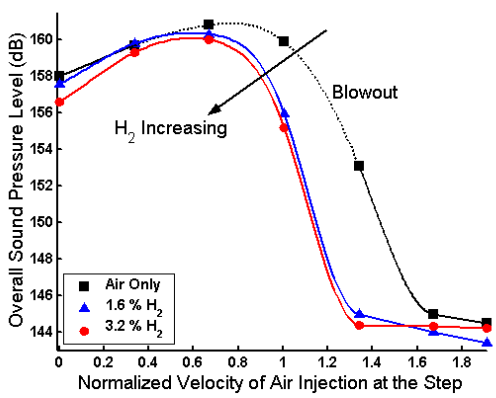


Fig. 7 Dependence of the Overall Sound Pressure Level on the flow rate of the static air injected near the step, for primary air flow velocity of 4.4 *m/s*, and a fixed fuel flow. The overall equivalence ratio ranged from 0.58 to 0.47 depending on the flow rate of the air jet and H2.

Figure 8 shows that the introduction of an air jet generally reduces NO levels. The data also show that the impact of the air jet may be non-monotonic when hydrogen is added to the primary fuel; only beyond the point of maximum pressure oscillations does a further increase of the air jet velocity guarantee substantial reduction in NOx. Before that, there is a complex dependence of NOx on the transverse jet velocity, mostly because of the competing effects of the dynamic flame structure, the equivalence ratio and the temperature fluctuations. Stable combustion, associated with both leaner and more aerodynamically compact flame in the presence of a transverse air jet, leads to a faster drop in NO. As in the case of pressure fluctuations, it appears that high transverse jet velocity is necessary to achieve a measurable reduction in NO formation. It is interesting to note that hydrogen addition under the same conditions boosts NO levels over the entire range, as the temperature rises in the combustion zone and in the products. Thus, if H2 is to be used to lower NOx, it must be supplemented with additional air, to be introduced strategically into the burning zone so as not to jeopardize the flame stability. In Ref [28], we were able to extend the flammability limit by using an air jet and H2 addition to stabilize the flame, all the way down to ϕ = 0.41. Combustion remained stable, with sub *ppm* NOx and 145 *dB* OASPL. This is ideal for applications requiring low NOx and stable operation without much product after treatment.

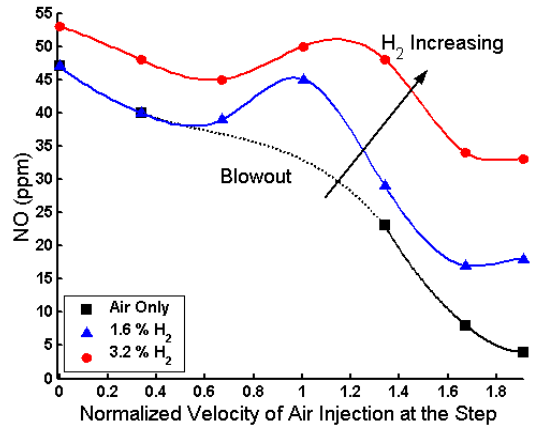


Fig. 8 NO level dependence on the velocity of the static air jet near the step for primary air flow velocity of 4.4 m/s, and a fixed fuel flow. The overall equivalence ratio ranged from 0.58 to 0.47 depending on the flow rate of air jet and H2.

4.2 Modulating the Air Jet (A_{air}) and H2 Injection (A_{H2a}) at the Step

Next, we investigate the impact of the transverse jet modulation on the pressure fluctuations. We show that one advantage of using modulation is to minimize the mass flow rate of the jet by controlling the phase using pressure feedback. We use a phase-shift controller to modulate the mass flow rate of the transverse jet. In these tests, the Reynolds number was fixed at 6300 with a nominal equivalence ratio of 0.57. The transverse jet to the primary air velocity ratios were 0.8, 1.25 and 1.7. Pure air, or an air-hydrogen mixture in which H2 contributed 1.6 % of the total heat addition, were used in the transverse jet. Hydrogen was premixed with air and injected through the slot. With air jet modulation, adding H2 to the transverse jet at the slot was more effective than adding hydrogen to the primary fuel. Paradoxically, without modulation, the opposite was true, i.e., adding H2 to the primary fuel stream was more effective.

To examine the response of combustion dynamics to the transverse jet modulation, we used a simple phase-shift controller with pressure feedback. A high pass filter with a cutoff-frequency at 5 Hz was incorporated to eliminate the drift in the pressure measurement and ensure that the duty cycle was always at 50%.

Figure 9 shows the OASPL as a function of $u_j/\overline{U}$ and the phase, with pure air injected at the slot. For $u_j/\overline{U}$ > 1.25 substantial reduction in OASPL could be achieved, but the performance is strongly dependent on the phase. Optimal phase angle in the phase-shift controller was found to be around 200 degrees. In the case with $u_j/\overline{U}$ = 1.25 the flame blew out at 60 and 120 degrees phase, as indicated by the dotted line in the Fig. 9. This is similar to the results observed previously where a small amount of cross jet was detrimental to stability and caused flame blowout. With $u_j/\overline{U}$ = 1.7 a pressure reduction up to 14dB is realized at the same phase angle of 200. Increasing the air jet velocity to $u_j/\overline{U}$ = 1.7 reduced pressure fluctuations throughout the entire range of phase angles, widened the stability band around 250 degrees and eliminated the flame blowout observed at lower velocities. It is interesting to observe that the maximum pressure reduction is comparable to that obtained with a static transverse air jet, showing some saturation effect. This was also observed in Fig. 7, where at high velocity ratios, the pressure level reached a plateau. Thus, regardless of the flow rate of the transverse jet, whether it is static or modulated, there appears to be limit on performance.

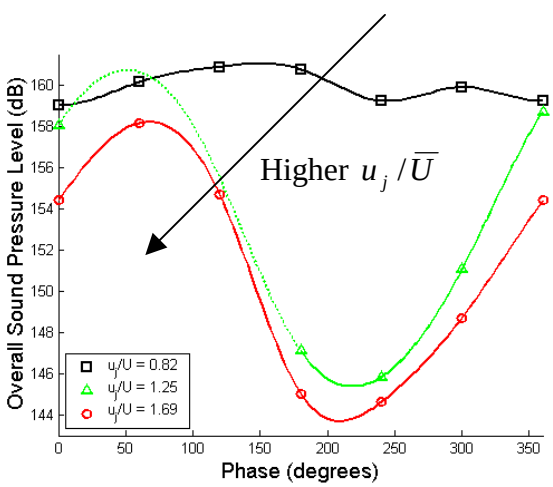


Fig. 9 Dependence of the Overall Sound Pressure Level on the flow rate of the modulated air jet without H2 addition injected near the step and the phase angle, for primary air flow rates of 4.4 *m/s*, and a fixed fuel flow. The overall equivalence ratio ranged from 0.52 to 0.48 depending on the flow rate of the air jet.

Adding H2 in the transverse jet significantly increases the effectiveness of control, as shown in Fig. 10. With H2 addition, the transverse jet velocity required for flame stabilization dropped by one half. By adding H2 at 1.6 % of the total heat release, the flame was stabilized at a velocity ratio of only 0.8, with a pressure drop up to 13 *dB* from the unstable case. Note that the optimal phase angle is now 300 degree. While a pure air jet reduces the burning velocity, the jet with H2 increases it, and we observed that H2 addition moves the burning zone closer to the step [28].

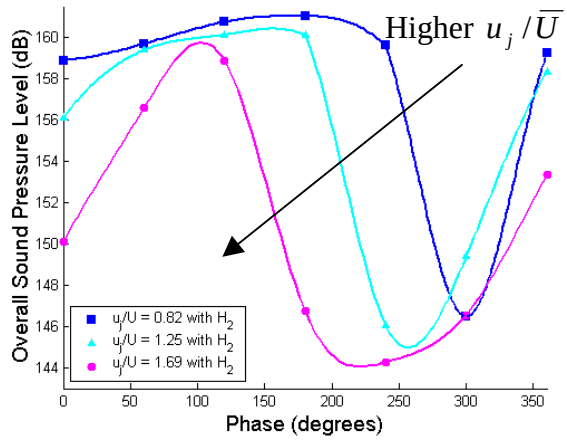


Fig. 10 Dependence of the Overall Sound Pressure Level on the flow rate of the modulated air jet with H2 addition injected near the step and the phase angle, for primary air flow rates of 4.4 *m/s*, and a fixed fuel flow rate. The overall equivalence ratio ranged from 0.52 to 0.49 depending on the flow rate of the air jet.

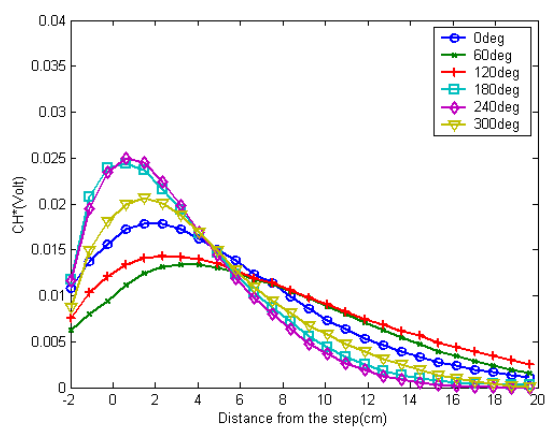


Fig. 11 Dependence of the mean CH* distributions on the phase angle of the controller for $u_j/\overline{U}$ = 1.7 with the overall equivalence ratio of 0.48 and without H2 addition.

Figure 11 shows the mean CH* distribution for the case of $u_j/\overline{U}$ = 1.7, over a range of the controller phase angles. Clearly, the burning characteristics change significantly with the phase-shift controller. At the optimal phase angles of 180 and 240 degrees, the burning zone is closest to the step and the burning rate

falls rapidly further downstream. In the most unstable case at 60 degree lag, burning extends much further downstream, the peak drops by one half and is located about 2 step heights downstream of the step.

5. Conclusion

In this paper, we address emission reduction in addition to instability suppression in a premixed backward-facing step combustor using a transverse air jet near the burning zone, and H2 addition to the air jet. We first develop an insight into the interactions between hydrodynamics, fuel/air mixture inhomogeneity and heat-release dynamics, and their impact on the instability of the combustion system. A parametric study was conducted to investigate the impact of flame-vortex interactions and fuel/air mixture inhomogeneity on the combustion instability. The location of the fuel injection was varied, which changed the total convective time delay and the level of fuel/air mixture inhomogeneity. Results show that when the primary fuel injection is moved further upstream, the impact of fuel/air mixture inhomogeneity is negligible and amplitude of pressure oscillations increase monotonically with increasing the Reynolds number. High speed CCD images of the flame confirm that a strong flame-vortex interaction is the dominant mechanism for heat release dynamics in all cases. For this reason, we focus the control effort on modifying the flame-vortex interactions in the combustion zone using a transverse jet of pure air or air-hydrogen mixture near the step.

We study experimentally the impact of a static and modulated transverse jet on stability and emissions. We demonstrate that with sufficient momentum, a static air jet injected perpendicular to the incoming flow near the step favorably modifies flame-vortex interactions and reduces pressure oscillations by up to 14 *dB*, while reducing NOx emission by an order of magnitude. We also investigate the impact of the transverse jet modulation using closed-loop control with pressure feedback. We show that modulating the air jet can achieve the same level of pressure reduction with 50% of the air flow rate required in the static air jet case, but the control effectiveness is strongly dependent on the phase angle. In both cases, adding small amounts of H2 leads to further reduction in OASPL.

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